

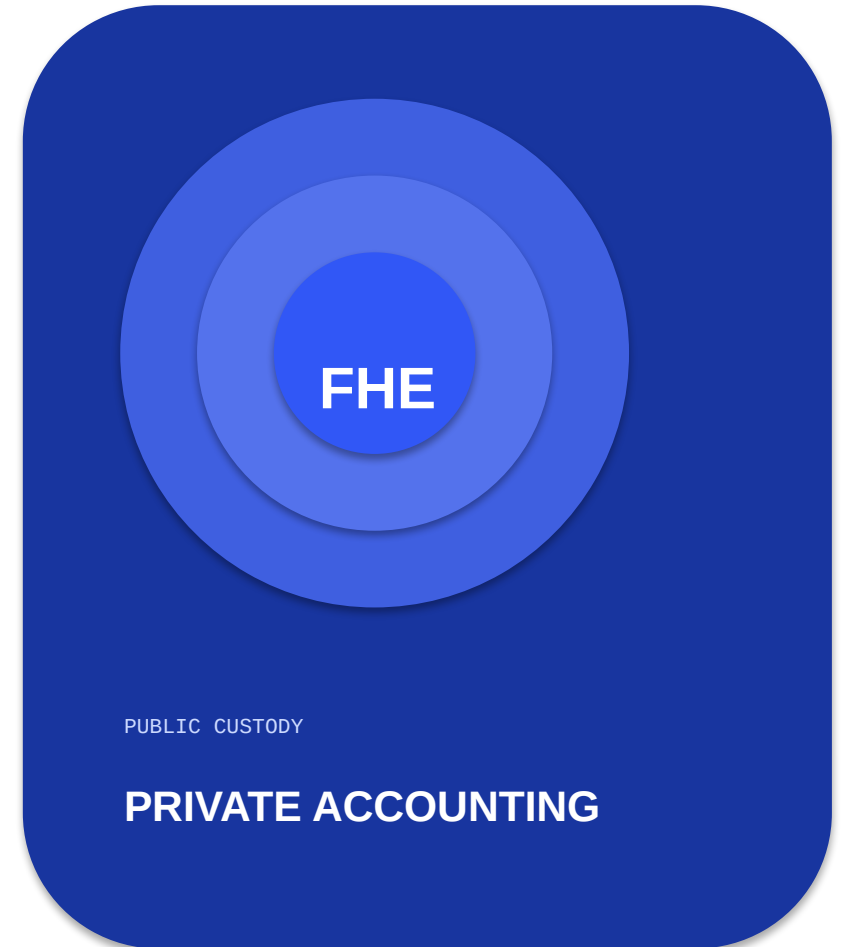
ZAMA FHEVM · CONFIDENTIAL FINANCE

Private prize savings, built for verification.

Principal stays withdrawable while encrypted balances and ticket weights stay confidential.

[Open the Veylott live demo](#)

RESEARCH SOFTWARE · NOT PRODUCTION



THE PROBLEM

Public ledgers expose private financial context

Auditability should not require publishing every saver's position.

01

Balance exposure

Wallet history can reveal a saver's position, deposits, withdrawals, and changing financial behavior.

02

Visible ticket weights

Transparent weights make whale activity easy to track and can disclose each participant's relative odds.

03

Timing reveals intent

Pending deposits and last-minute entries expose behavior around a prize round before it settles.

THE SOLUTION

Public custody. Confidential accounting.

Veylott separates what must be auditable from what should remain private.

PUBLIC · INDEPENDENTLY VERIFIABLE

Custody layer

- Deposited testnet cUSDC
- Contract and custody bindings
- Transaction receipts
- Aggregate solvency

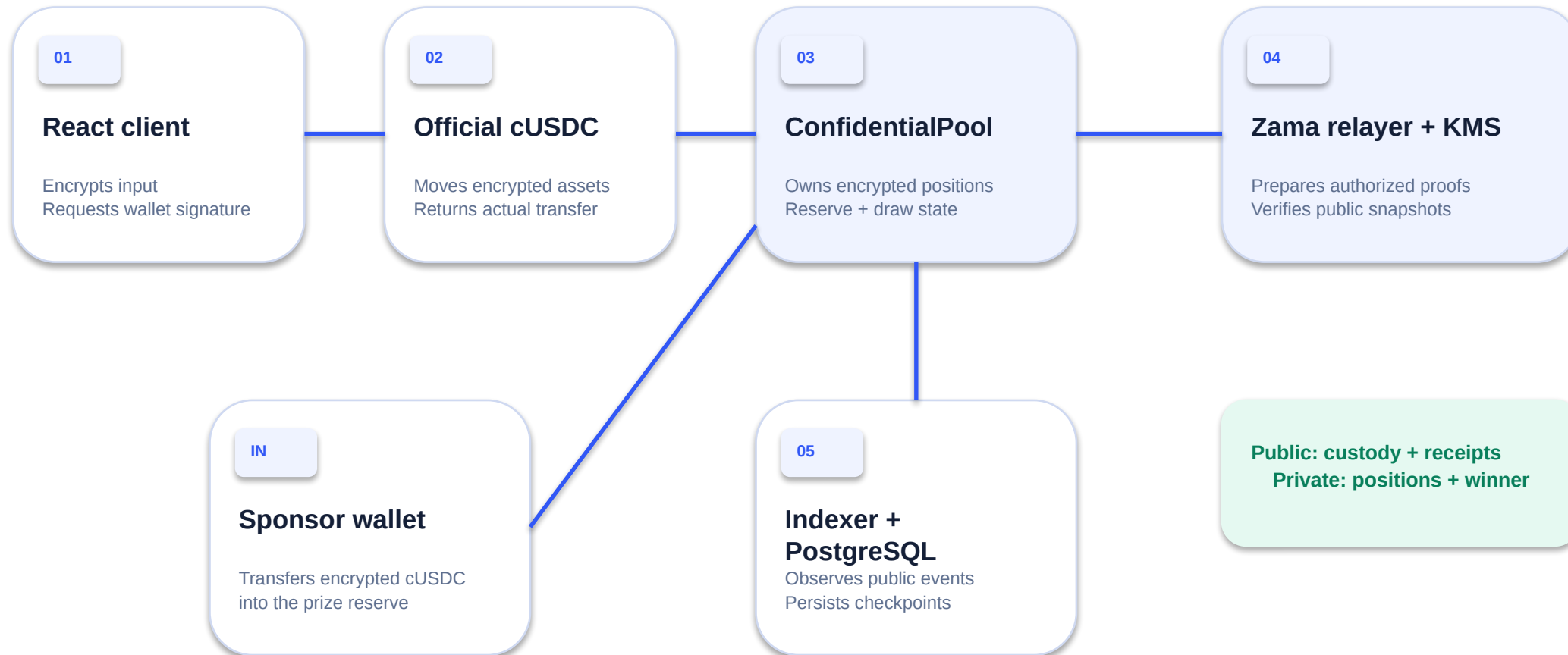
ENCRYPTED · WALLET-AUTHORIZED

Privacy layer

- Running balances as euint64
- Weighted draw intervals
- Prize credits
- Local balance reveal

ARCHITECTURE

A verifiable path from wallet to confidential settlement



LIVE PRODUCT

The console is live—and honest about connection state

PRIVATE PRIZE SAVINGS

Save privately

Prize rounds

How it works

Connect wallet

CONFIDENTIAL SAVINGS CONSOLE

Your position stays private.

Deposit confidential cUSDC, enter verifiable prize rounds, and withdraw without publishing amounts, balances, or ticket weight.

Encrypted accounting

Wallet-bound access

Verifiable exits

RUNTIME ASSURANCE

PENDING

Protocol state

Network

Indexer

Transactions

Wallet Disconnected

Online

Safety-Locked

Writes remain locked until runtime verification succeeds. Connect a Sepolia wallet to verify the active deployment.

Indexer Online

Pool Checking

Wallet Disconnected

Deployment Pending

Checked 10:07:50 PM

VERIFIED ELIGIBLE WEIGHT

AWAITING ROUND

New pool; appears after the first finalized draw

VERIFIED PRIZE RESERVE

AWAITING ROUND

New pool; appears after the first finalized draw

PRIVATE SAVERS

LOADING

—

Loading verified source...

CONFIRMED ROUNDS

LIVE

0

No finalized rounds yet

PRIVATE OPERATIONS

Manage your savings

Balances reveal only after wallet authorization. Deposits and settlement remain independently verifiable on-chain.

PRIVATE POSITION

ACL PROTECTED

Only you can reveal this balance

The ciphertext stays on-chain. Decryption happens locally after wallet authorization.

AVAILABLE PRINCIPAL

●

●

●

●

●

Authorize a one-time EIP-712 request to decrypt the position assigned to your wallet.

STEP 01

ENCRYPTED END-TO-END

Fund your private position

The amount transfers as cUSDC and remains encrypted in calldata, events, and pool accounting.

Amount

0.00

Wallet balance is private

cUSDCMock

STEP 02

ONE TRANSACTION

Withdraw confidentially

The requested amount and transferred cUSDC remain encrypted throughout settlement.

Amount

0.00

Encrypted sufficiency check

cUSDCMock

ERC-7984 transfers zero if your private balance is insufficient. Reveal your position afterward to verify

PRODUCTION CAPTURE

Clear wallet state

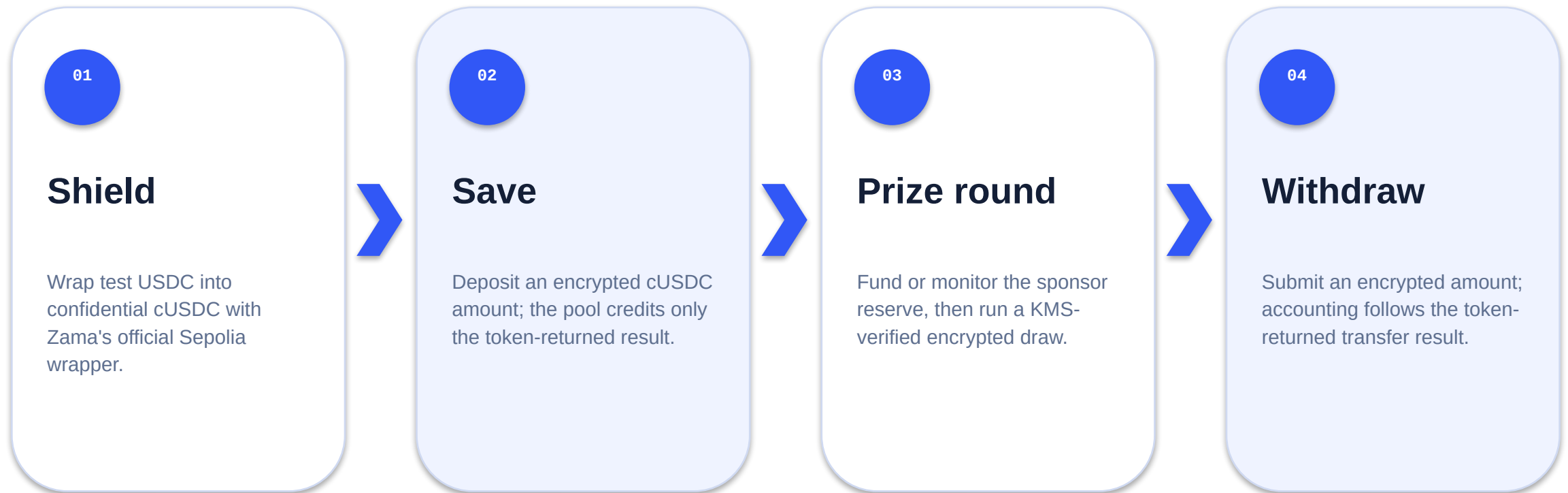
A disconnected user sees only Connect wallet. An address appears only after the provider returns the selected account.

Runtime assurance

Chain, bytecode, and custody bindings are checked before protocol writes.

Captured from the Veylott live demo · 4 Sep 2026

Four explicit steps. No hidden signing.



SECURITY CONTROLS

Solvency and settlement invariants are enforced in code

01

Asset-bound credit

Encrypted deposit credit is derived from the token-returned custody amount.

02

Reserved prizes

Each draw consumes verified sponsor funds so assets cannot fund repeated awards.

03

Complete liabilities

Winner credit enters both the encrypted position and aggregate at finalization.

04

Bound draw proofs

KMS evidence must match the stored aggregate and reserve handles.

05

Timeout recovery

Anyone can release a stale draw lock after the 24-hour cancellation delay.

VERIFIED EVIDENCE

A completed historical prize round—not a simulated receipt

Each transaction below links directly to its confirmed predecessor-deployment receipt.

01

cUSDC deposit

[0xe36db7...b39f](#)

Encrypted position

02

KMS draw

[0x504862...ce6c](#)

Draw 1 finalized

03

Private claim

[0x5763be...c969](#)

Ordinary withdrawal

Deposit → KMS draw → private claim → principal withdrawal · all confirmed

Reproducible tests—and durable live infrastructure

FULL

VALIDATION SUITE

Foundry contracts · backend API/indexer
client encryption · frontend UX

Vercel

React frontend

PRODUCTION UI

Render

Node indexer/API

PUBLIC API

Neon

PostgreSQL checkpoint

RESTART RECOVERY

Indexer checkpoint restored successfully after a live service restart

OPEN RESEARCH · VERIFIABLE EVIDENCE

Confidential savings, without blind trust.

Try the live app. Inspect the active Sepolia deployment. Review every invariant and test.

[Open the Veylott live demo](#)

github.com/Webghost01-NG/veylott

ACTIVE POOL

0x54FdC46D
0EA722Ef
A4853192
678b35fC
ABFad99C

ETHEREUM SEPOLIA